1 The nature of technology: Robots



You will learn

- what a robot is and how it works
- what control systems can and cannot do
- what robots can do now and how they are developing.

People have been fascinated by robots for hundreds of years. Fictional robots appear in science fiction films such as *Star Wars*. Robots appear in books and in video games. Today, robots are with us in real life. Robots are used to make electronics equipment such as TVs and computers. They are used to build cars. Robots drive cars. We live in an age of robots.



Talk about...

Some robots are designed to look like people. Do you think that makes them more likeable? Or do you prefer robots that don't look like people? Think of examples of robots in films and TV shows. Which ones do you like?

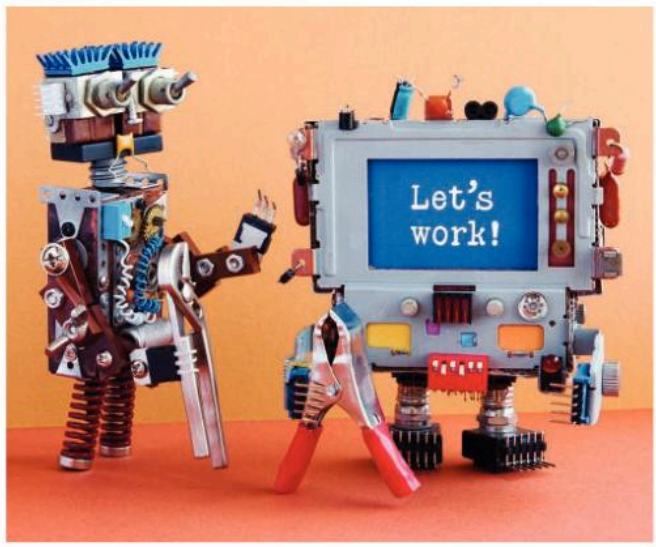


Learning outcomes: Describe the technology of robotics and control systems; Discuss the potential and the limitations of robotics and control systems

Class activity

Design your perfect robot assistant. What will your robot look like? What special tools and powers will it have? What jobs will your robot assistant do that will make your life easier?

robot robot arm drone
actuator sensor
controller control system
artificial intelligence
robot car nanobot



Did you know?

A British scientist called Kevin Warwick had a computer chip implanted into the nerves of his arm. In an experiment that took place in 2002, he controlled a robot hand over the internet. As he moved his hand in the USA, a robot arm in the UK made the same movements. In the future, people who have lost limbs might have mechanical replacements that work just as well as the real thing.



1.1 What is a robot?

In this lesson

You will learn:

- what a robot is and what it looks like.

Spiral back

In Student Book 4, you learned that microprocessors are built into machines and devices that we use every day. Cars, refrigerators and TVs have built-in microprocessors. A microprocessor adds more power to a device and makes it easier to use.

One machine is made more powerful and useful by microprocessors than any other. That machine is the robot. In this unit, you will learn how robots work and what they do.

What is a robot?

A **robot** is a machine. Robots are built by people to do a job. A robot must be programmed by a human to do its job. Once a robot is programmed, it does its job without help from humans.

Unless it breaks down, a robot does its job automatically. It does not need a human to control its actions. A robot does not get bored with its job. It does not make mistakes. The only time a human is needed is if the robot develops a fault.

What does a robot look like?

When people think of a robot, they often imagine them to look like people. Robots in science fiction films are often shown with arms, legs, a body and a head. In the real world, it is different.



Humanoid robots

Humanoid means like a human. Some robots are made to look like people. The robots take part in competitions where they carry out day-to-day tasks just like humans would. They are set challenges such as climbing a flight of stairs and carrying drinks without spilling them.

These robots are used by scientists to develop methods that might be used in the future.

Robot arms

Most robots are built to do jobs that humans do. Humans use their hands and arms to do most jobs. To do the same jobs, a robot has to be able to move like a human. This is the reason that many robots look like a human arm.

A **robot arm** has joints that bend and twist. A robot arm has a hand to grip things with.



Non-humanoid robots

Many robots do a job that humans do without looking human. An example is a robot vacuum cleaner. Another example is a **robot car**. A device does not have to look like a human to be called a robot.



Activity

Invent your own robot which would help with cleaning. Think of a name for your robot. List the main functions of your robot and the advantages of using it to do the cleaning.



Extra challenge

Design a poster to advertise the new robot cleaner.

Be creative

Create an image of the robot cleaner and include it in your poster.

Think again

Is your robot cleaner humanoid or not? Why did you make that choice?

1.2 What robots are used for

In this lesson

You will learn:

- what kinds of jobs robots can do
- how robots are used to build cars.

What jobs can robots do?

Repetitive tasks

Robots do jobs that are repetitive. This means jobs where an action needs to be done in the same way many times. An example is bolting wheels onto a car. Eight thousand wheels are put onto cars every day in a typical car factory. Humans find this kind of work boring. They can get tired and make mistakes or injure themselves. Robots never get bored or make mistakes.

Detailed work

Some jobs involve tasks to be carried out with great accuracy. A single mistake can mean a whole product is spoiled and must be thrown away. Making microprocessors involves a lot of detailed and accurate work. Robots are used to make microprocessors and other electronic equipment such as TVs.

Dangerous work

Robots are used in places where it is dangerous for humans to work. For example, robots are used on space missions. Robots called Sojourner and Pathfinder were sent to explore the surface of Mars. The picture gives you an idea of what Sojourner looked like.

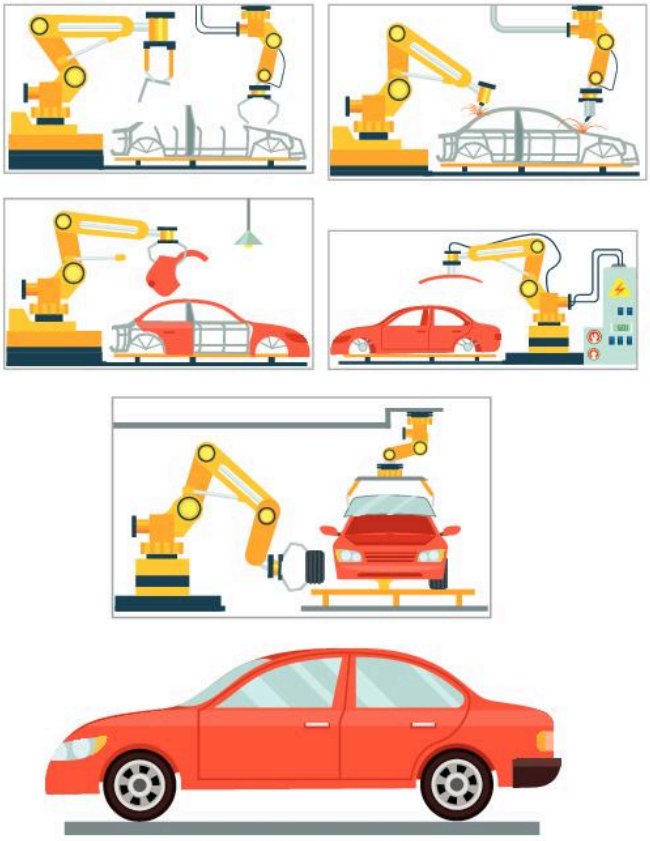


Robots are used under the sea. They explore the sea bed or inspect and repair oil rigs. Robots also work in environments where chemicals or radioactivity make it unsafe for humans to work.

Below are the main differences between how robots and humans do jobs.

Robots	Humans
• More accurate than humans	• Intelligent – they can find better ways of doing jobs
• Do not get bored or tired	• Creative – they can invent new things, such as robots
• Work faster than humans	• Solve problems
• Do not make mistakes	• Have emotions and feelings

Robots in the car industry



A car is not built by a single robot. The car moves along an assembly line. The car stops at points along the line. At each point, robots work together to build part of the car. For example, one team of robots will build the engine. Another team will put on the doors.

Robots must work together to make sure every task is performed in the right order. They must avoid colliding with each other. These teams are called **collaborative robots**. Collaborate means to work together.

Painting

Robots are used to paint cars. Robots always give cars a high-quality finish. The paints used on cars are hazardous to humans. Humans do not have to risk their health to do the job.



Activity

Make a poster with the title 'Robots versus humans'. List the strengths of robots and humans in your poster. Illustrate your poster.



Extra challenge

Describe a real-life example where robots are used to do dangerous jobs. Research the web to find your example. Some ideas for your research are fire-fighting, clearing landmines and inspecting pipelines. Are there images or videos of the robots in action?

Think again

Research how robots are used in the car industry. Find examples of jobs that robots are good at. Find examples of jobs that are still done by humans. Write up your findings in a report.



Digital citizen of the future

Some researchers think robots will replace 800 million jobs worldwide by 2030. Workers will need to train to do new jobs. Many of these jobs will be in the robotics industry. Having up-to-date computer skills will be more important than ever. Will you be ready for the new world of work?

1.3 How robots work

In this lesson

You will learn:

- how a robot uses sensors, a controller and actuators to do its work.

What makes a robot work?

You must balance a tennis ball on a book and walk across the room without dropping the ball. How do you do this?

As you cross the room, you use your senses to warn you when the ball is starting to roll. Your brain works out how to stop the ball from falling. Your brain tells your muscles how to move your arm to stop the ball rolling.

A robot works in the same way.

- A robot has **sensors** that act like your eyes and ears. They tell the robot what is happening in the world around it.
- A **controller** is the brain of a robot.
- **Actuators** are the muscles of a robot. They make the robot move.

Sensors

A robot does not have eyes and ears like humans. It uses components called sensors.

- Video cameras give the robot sight.
- **Proximity sensors** tell a robot how close an object is.
- **Bumpers** tell the robot it has collided with something.
- **Pressure sensors** let a robot know that it is touching something.
- Sensors that detect chemicals give a robot its sense of smell and taste.

A robot uses the sensors it needs to do its job.



Controller

A controller is the brain of a robot. The controller is programmed to do a job. A robot cannot think like a human. It carries out whatever instructions it has been programmed to do.

Actuators

A robot does not have muscles and bones like humans. It uses mechanical parts to lift and move objects. The mechanical parts that make a robot move are called actuators. The actuators in a robot do the same job as the muscles in your arm.

How the robot vacuum cleaner works

A robot vacuum cleaner helps to clean a house. A motor is the only extra mechanical part that a robot vacuum cleaner has compared to a normal cleaner. The motor moves the cleaner around the home on wheels.

Proximity sensors measure a room so that the cleaner can calculate the best route to take. Proximity sensors use a light beam to calculate distance. When the light beam hits something solid, it bounces back to the sensor. The time it takes to bounce back tells the sensor how far the light travelled.



A sensor called a bumper detects if the cleaner hits a wall or piece of furniture. If the cleaner detects a collision, it follows an algorithm. The cleaner makes a small turn and tries to move again. Once it finds a clear path it continues cleaning.

The cleaner has a **vibration sensor** on the base. If a lot of dirt hits the sensor, the vibration it causes tells the cleaner that area needs more cleaning.

A sensor warns the cleaner if the battery power is low. The cleaner finds an electrical charging unit using proximity sensors. The cleaner recharges its batteries before continuing cleaning.



Activity

Give two examples of the sensors a robot vacuum cleaner uses. How are the sensors used?

Think again

What is an actuator? Why does a robot need actuators?



Extra challenge

Have you seen a 'barista' in a coffee shop? A barista is the person who makes the cups of coffee. Could a robot be made that works as a barista? Describe how sensors and actuators might be used by a robot barista.

1.4 What is a control system?

In this lesson

You will learn:

- what control systems are used for
- what a control loop does.



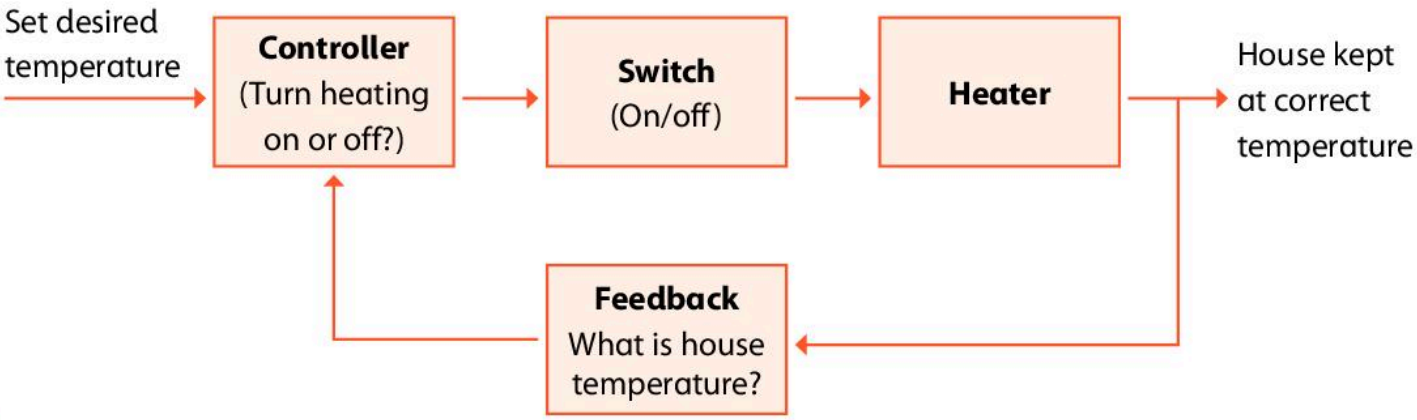
What is a control loop?

If you have a heating system in your home, you may have seen a controller like this:

A heating controller lets you set an ideal temperature for your home. Suppose your ideal temperature is 20 degrees Celsius.

- When the heating is switched on, the controller sends a message to turn on the radiators.
- The message passes through a **temperature sensor**.
- The sensor sends a message back to the controller telling it the actual temperature in the house.
- When the temperature reaches 20 degrees, the controller turns off the radiators.
- The temperature sensor keeps sending messages back to the controller. When the temperature falls below 20 degrees, the radiators are turned back on.

This is a **control loop**. It is sometimes called a **feedback loop**. The sensor feeds back information to the controller.



Control systems in the home



Control systems are installed in many homes. They control temperature, lighting and security.

Modern homes may have many sensors installed:

- Heating systems use temperature sensors.
- Alarm systems use proximity sensors and pressure pads to detect intruders.
- Security systems use video cameras.
- Lighting systems use proximity sensors to detect movement.

Lights can be turned on automatically when a sensor detects someone entering a room. They are turned off again when the sensor detects the room is unused. Heating can be kept at a lower level when the house is empty and warmer when people are in it.

The controller is connected to the internet, so the house owner can control systems from anywhere. Heating can be turned on so the house is warm when people arrive home.

Devices such as ovens can be turned on so food is cooked when people arrive home from work. Messages are sent to the home owner if an intruder is detected. Security cameras can be viewed on a smartphone. The home owner can even view the contents of their fridge to find out what they need to buy from the supermarket.

A house with control systems is called a **smart home**. Control systems make life easier, more secure and reduce energy use.



Activity

You are travelling home in your robot car with your parents. What messages can you send to your smart home so that it is welcoming when you arrive?

Think again

A feedback loop can be used to control air conditioning to keep a home at a low temperature. Draw the feedback loop (or control loop) for this process.



Extra challenge

Use your program design skills to write an algorithm. The algorithm should describe how a heating controller keeps the temperature in the house at 18 degrees Celsius.

1.5 Living with robots

In this lesson

You will learn:

- how robots are changing the world you live in.

You have learned how robots help to make things in manufacturing industries. Robots and control systems make life easier in homes. Robots are used in many other industries.

Robots on farms

Harvesters, ploughs and other farm machines are driven by robots. The vehicles find their way around fields using the same **satellite navigation** (satnav) that car drivers use. The farmer does not have to drive farm vehicles so has time to do other tasks on the farm.

A **drone** is a helicopter or plane that does not need a pilot. Drones are robot-controlled. Drones are used to spray pesticides or fertilisers where they are needed.



Robot arms are used to harvest and pack produce. Robots use sensors to decide what is ripe. Robot arms with pressure sensors can pick delicate fruits and vegetables without causing damage.

Shopping with robots

More people buy products online. Online shops store the products in large warehouses. Robots find and pack items when they are purchased.

Online stores are experimenting with using robot vehicles and drones to deliver goods. Some high street stores are using robot shop assistants. They move around the shop and answer any questions that customers have.



Robots and jobs

Some people worry that robots will replace jobs and cause unemployment.

Other people say that robots and technology will replace some jobs but will also create new ones.

- Robots are good at repetitive tasks that need to be done quickly and accurately. Many of the jobs that robots replace are low-skilled. People often find the jobs stressful and boring.
- People are needed to invent and build new robots. New factories are needed for robot-controlled machines. **Computer technicians** are needed to install robots. **Software engineers** write programs to control robots. Technicians are needed to maintain and repair robots. The new jobs are often high-skilled and interesting.

Some people say introducing robots will create more new jobs than are lost.



Activity

Have a look at the two car assembly lines. One is from the 1930s when humans did all the work. The other is a modern assembly line with robots. Describe the differences you can see between the two pictures.





Think again

Which of the two assembly lines would you prefer to work in? Explain why.



Extra challenge

Do research on robots working in agriculture or retail. What kind of robot vehicles and machines are used in these industries?

1.6 Robots of the future

In this lesson

You will learn:

- how robots will develop in the future
- how robots are helping doctors
- how robots will change the way we travel



Artificial intelligence

Robots are being developed that can act more like humans. This is called **artificial intelligence** (AI). AI allows robots to:

- recognise pictures and speech
- learn better ways of doing things
- decide the best way to deal with situations.

In the future, a robot shop assistant may recognise how a customer feels from their voice and face. The robot will take a different approach depending on how the customer looks and sounds.

In the classroom, a robot assistant will learn about your progress and how you study. The assistant will set you individual projects and find learning materials that are best for you.

Robot vehicles



Robot cars are also called autonomous cars. Autonomous means the car works automatically with no need for any human help. An autonomous car does not need a driver. It has sensors that can detect pedestrians, other cars and road signs. The controller uses the information from the sensors to operate the accelerator, brakes and steering.

At any moment there will be different ways a robot car can deal with a situation on the road. For example, does it brake or swerve to avoid a collision? AI will help the car make the best decision.

Cars in the future will look different to cars today. An autonomous car does not need a steering wheel or pedals. Seats do not all need to face forward.

Robots in healthcare



Robots are already being used in modern hospitals. Scanners create a 3D picture of the inside of our bodies so that doctors can diagnose problems. Robots help doctors carry out delicate surgery they could not do without robotic instruments. Robots help doctors but will not replace them.

Robot nursing assistants carry out basic health checks on patients. These robots already help look after elderly patients who need regular care. A robot called Robear can lift patients in and out of their bed or wheelchair. It is more comfortable for the patient and prevents nurses from being injured.

Scientists are researching **nanobots**. A nanobot is a microscopic robot. In the future, nanobots may be able to work inside a patient's bloodstream to fight infections or carry out delicate surgery.



Activity

Travelling in cars is one of the most dangerous things we do. Will autonomous cars make car travel safer or more dangerous? Discuss this in a small group, then write down your own opinions.



Explore more

Choose one use for robots you have learned about in this unit. For example, robots in the home or robot cars. For the area you have chosen, find:

- examples of how robots are used
- photographs of robots
- interesting facts and figures.

Go back over the content of this unit and do extra research using the web.

Present your findings as a presentation, word-processed document or website.



Extra challenge

Carry out more research on either robot cars or robots in healthcare. Make notes. Find examples, facts and pictures.



Check what you know

You have learned

- what a robot is and how it works
- what control systems can and cannot do
- what robots can do now and how they are developing.



Activities

You have been asked to design a robot. The robot should throw tennis balls to a student to help them practise catching. The robot needs to throw the balls safely.

1 Draw the robot. Mark at least one part of the robot used for sensing. Mark at least one part used for moving.

2 Describe the sensors your robot needs to decide where and how far to throw the ball. Describe how the robot might throw the ball.

3 It would be useful for the robot to know if the student has caught the ball. What type of sensor would it need? There is more than one right answer to this question.

If you have finished the activities above and want an extra challenge, try these:

4 The robot could throw the ball to different places – sometimes to the left, sometimes to the right, and at different heights. But not too far away so the catch is impossible. How could you change the robot to work in this way?



5 A robot could use AI to develop the catcher's skills. The robot will learn where the catcher finds it easy or difficult to catch a ball. For example, if the robot learns the catcher finds high balls or balls to the left difficult, it can give the catcher practice with these types of throws. What sensors can you use so the robot knows if the ball has been caught? Write an algorithm to explain how your robot will work.



Test

1. What is a robot?
2. Give an example of a control system that might be used in someone's house.
3. What are proximity sensors? Why might a robot use proximity sensors?
4. Why are robots better at doing repetitive tasks than humans?
5. Describe something you have found out from your own research about robots – either how they are used now or might be used in the future.



Self-evaluation

- I answered test questions 1 and 2.
- I completed activity 1.
- I answered test questions 1–4.
- I completed activities 1 and 2.
- I answered all the test questions.
- I completed activities 1–3.

Re-read any parts of the unit you feel unsure about. Try the test and activities again – can you do more this time?